

Passive acoustic gliders as a complement to ship-based marine mammal surveys

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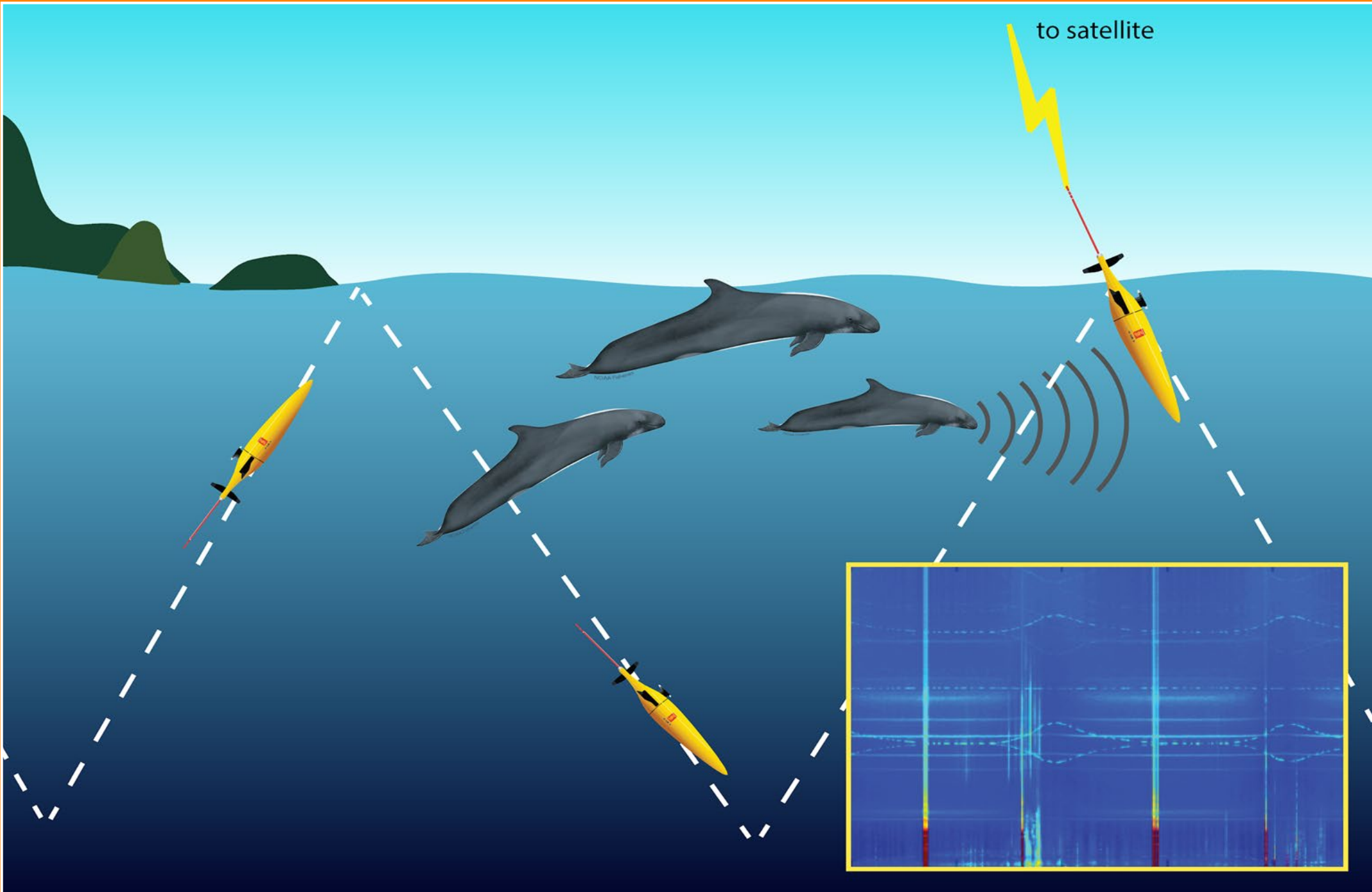
Marine Mammal Institute, Oregon State University | NOAA Fisheries, Pacific Islands Fisheries Science Center | NOAA Fisheries, Southwest Fisheries Science Center | Cooperative Institute for Marine Ecosystems and Resources Studies, Oregon State University and NOAA Pacific Marine Environmental Laboratory (PMEL) | Saltwater, Inc. in support of NOAA Pacific Islands Fisheries Science Center | University of California, Santa Cruz, Institute of Marine Sciences' Fisheries Collaborative Program



Background

- NOAA Fisheries is mandated by Congress to monitor marine mammals to mitigate current risks and support sustainable oceans that include marine mammals as functioning parts of the marine ecosystem
- Ship-based visual and towed array acoustic surveys have traditionally been the “gold standard” for cetacean surveys, but conducting regular ship-based surveys is becoming more difficult due to shortages of adequate ship time

- Gliders equipped with passive acoustic monitoring (PAM) systems offer an effective, efficient alternative survey modality that can augment existing surveys and improve our understanding of marine mammals to support their long-term conservation
- The U.S. West Coast and Pacific Islands Regions cover the largest area under NMFS jurisdiction and have existing glider assets and expertise so are working together to accelerate PAM glider operations across the Pacific

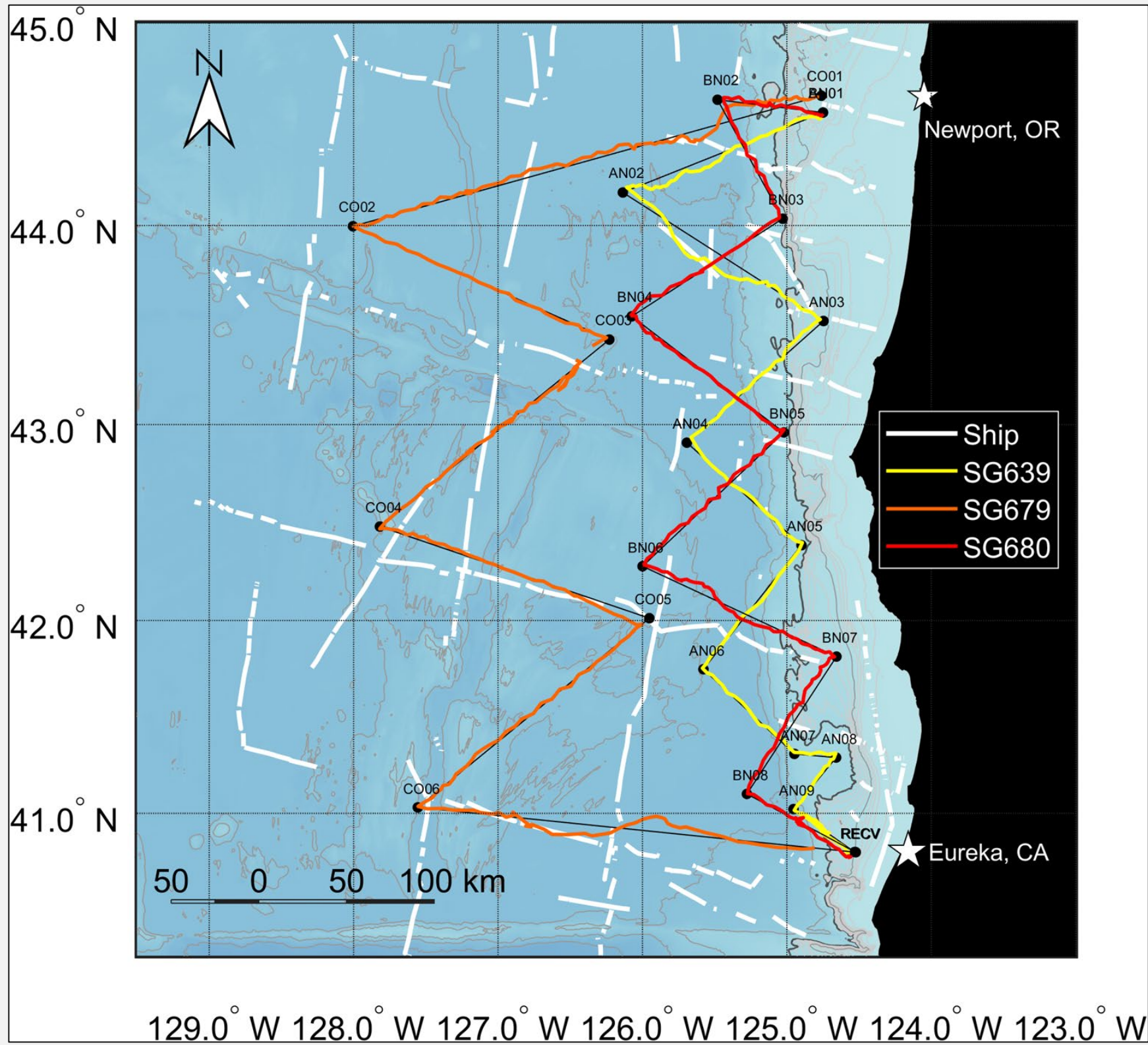


PAM gliders can record near continuous high-frequency acoustic data allowing researchers to record the calls of most marine mammal species. The spectrogram (above right corner) shows 12 hours of acoustic data with a few bouts of glider VBD pumping (red lines) and echolocation clicks and whistles of false killer whales (light blue to yellow vertical lines and cloud)

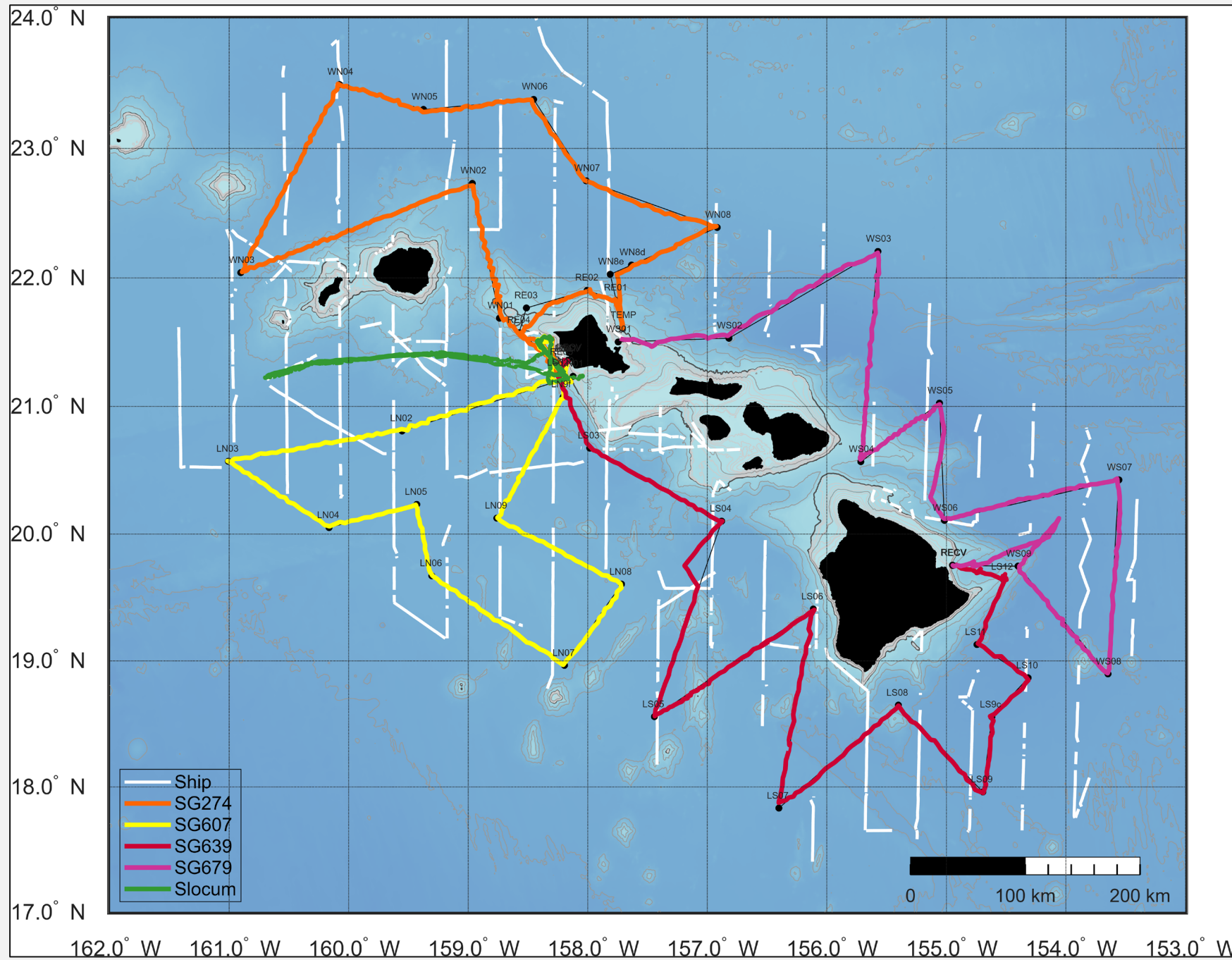
Concurrent PAM glider and ship-based surveys

- A key step towards fully utilizing PAM gliders for marine mammal monitoring is understanding how glider-based acoustic observations compare to visual sightings and acoustic detections made during conventional shipboard surveys
- To support this effort, we conducted multiple glider missions that coincided with large-scale ship-based marine mammal surveys
- Three Seaglidgers surveyed off the coast of Oregon and Northern California in fall 2024, overlapping in time and space with the California Current Cetacean and Ecosystem Assessment Survey (CalCurCEAS). Gliders collected acoustic survey data over 3000 km and 145 survey days and collected CTD profiles for 746 dives
- Four Seaglidgers and one Slocum glider surveyed around the Main Hawaiian Islands in winter 2026 at the same time as the Winter Hawaiian Islands Cetacean and Ecosystem Assessment Survey (WHICEAS). Gliders collected acoustic data over 4100 km and 222 survey days and collected CTD profiles for over 1000 dives

CalCurCEAS 2024



WHICEAS 2026



Maps show the extent of gliders (colored lines) and ship survey (white lines) for CalCurCEAS 2024 (left) and WHICEAS 2026 (right)

Conclusions

- Gliders equipped with passive acoustic recorders allow us to increase survey effort (distance and time) and cetacean encounter rates which provides more robust data on cetacean presence and absence
- These concurrent surveys provide a unique opportunity for side-by-side comparison of ship and glider-based cetacean surveys. Analysis of acoustic presence of sperm whales, baleen whales, and beaked whales is in progress
- Work is ongoing to integrate PAM glider data into species distribution models and stock assessments and to further develop the statistical framework for population density estimation from slow-moving platforms
- NOAA Fisheries is developing a research-to-operations plan to serve as a roadmap for sustained PAM glider cetacean surveys



Acknowledgments

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Contact

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